

Finding Quadratic Equations and Using the Discriminant to Count X-Intercepts

Using the Discriminant to Find the Number of X-Intercepts of a Parabola

Review

Standard Form of a Quadratic Equation

$$ax^2 + bx + c = 0$$

The solution to all quadratic equations in standard form.

The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The Discriminant

$$D = b^2 - 4ac$$

The discriminant gives us information about the number of solutions of a quadratic equation and the type of solutions because it lies under the root symbol in the quadratic formula.

I) If the discriminant of a quadratic equation is **positive**, then the equation has **two real solutions**.

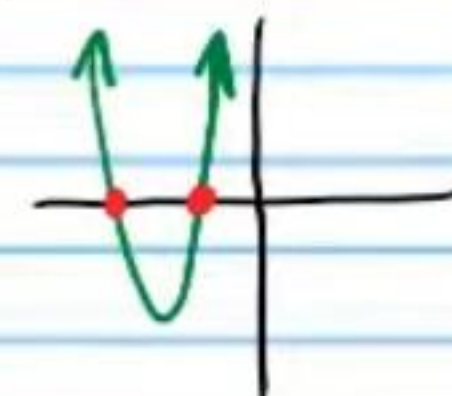
II) If the discriminant of a quadratic equation is **zero**, then the equation has **one real solution**.

III) If the discriminant of a quadratic equation is **negative**, then the equation has **two complex solutions** or we can say that there are no solutions.

Using the Discriminant to Find the Number of X-Intercepts of a Parabola
If $ax^2 + bx + c = 0$, then the discriminant, $b^2 - 4ac$, can tell us the number of x-intercepts that the parabola has which is generated by the corresponding equation $ax^2 + bx + c = y$.

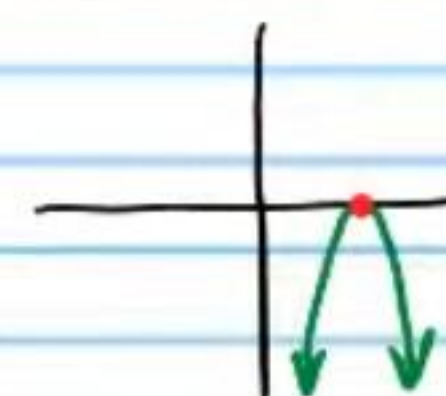
Examples

I) $b^2 - 4ac > 0$



Two X-intercepts

II) $b^2 - 4ac = 0$



One X-intercept

III) $b^2 - 4ac < 0$



Zero X-intercepts

Use the discriminant to find the number of x-intercepts of the graph of each quadratic expression below.

① $y = -2x^2 - 5x + 1$

$$0 = -2x^2 - 5x + 1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$D = b^2 - 4ac$$

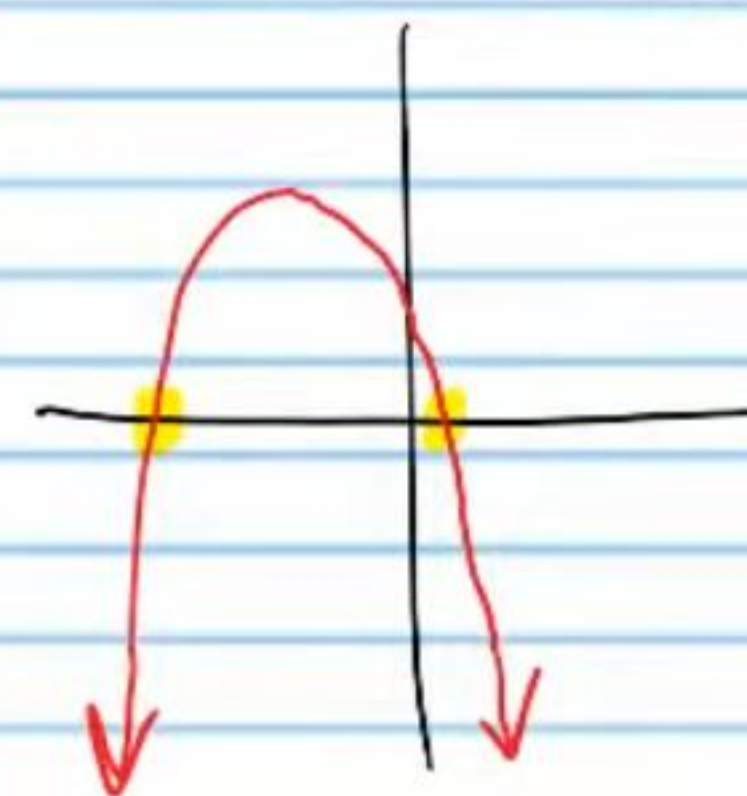
$$D = (-5)^2 - 4(-2)(1)$$

$$D = 25 + 8$$

$$D = 33$$

Two Solutions

Two X-intercepts



$$\textcircled{2} \ x^2 + 12x + 36 = y$$

$$x^2 + 12x + 36 = 0$$

$$D = b^2 - 4ac$$

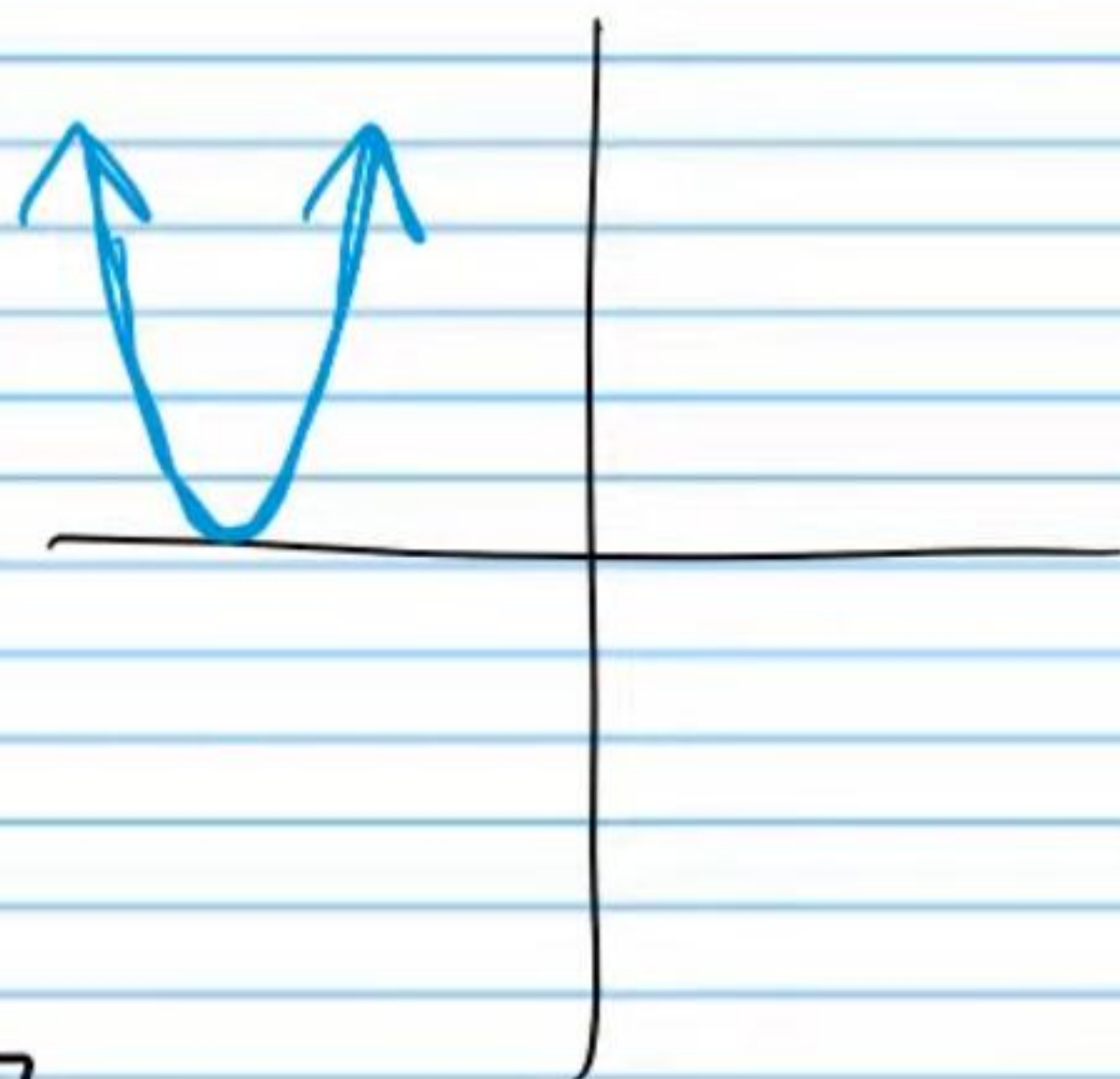
$$D = 12^2 - 4(1)(36)$$

$$D = 144 - 144$$

$$D = 0$$

One Solution

One X-intercept



$$\textcircled{3} \ y = -3x^2 + 4x - 2$$

$$0 = -3x^2 + 4x - 2$$

$$D = b^2 - 4ac$$

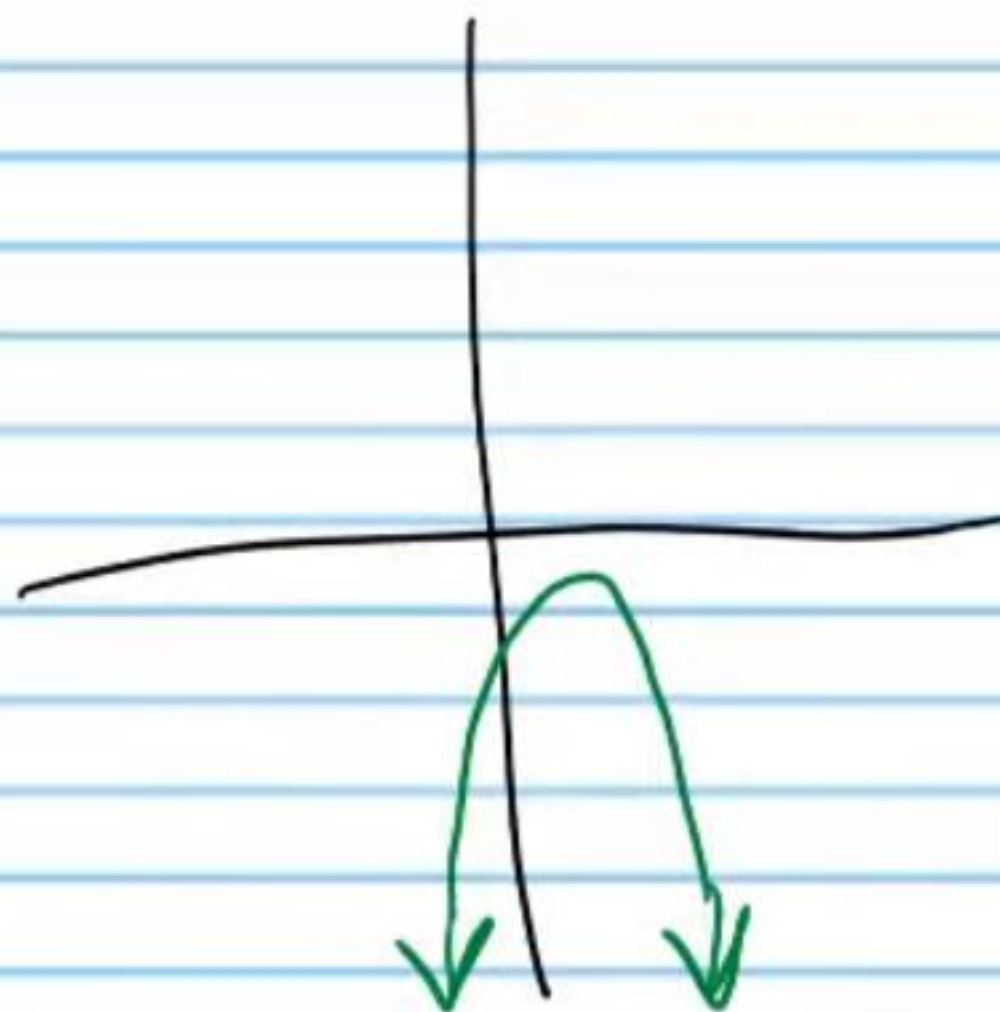
$$D = 4^2 - 4(-3)(-2)$$

$$D = 16 - 24$$

$$D = -8$$

No Solutions

No X-intercepts



$$\textcircled{4} \ y = 25x^2 - 10x + 1$$

$$0 = 25x^2 - 10x + 1$$

$$D = b^2 - 4ac$$

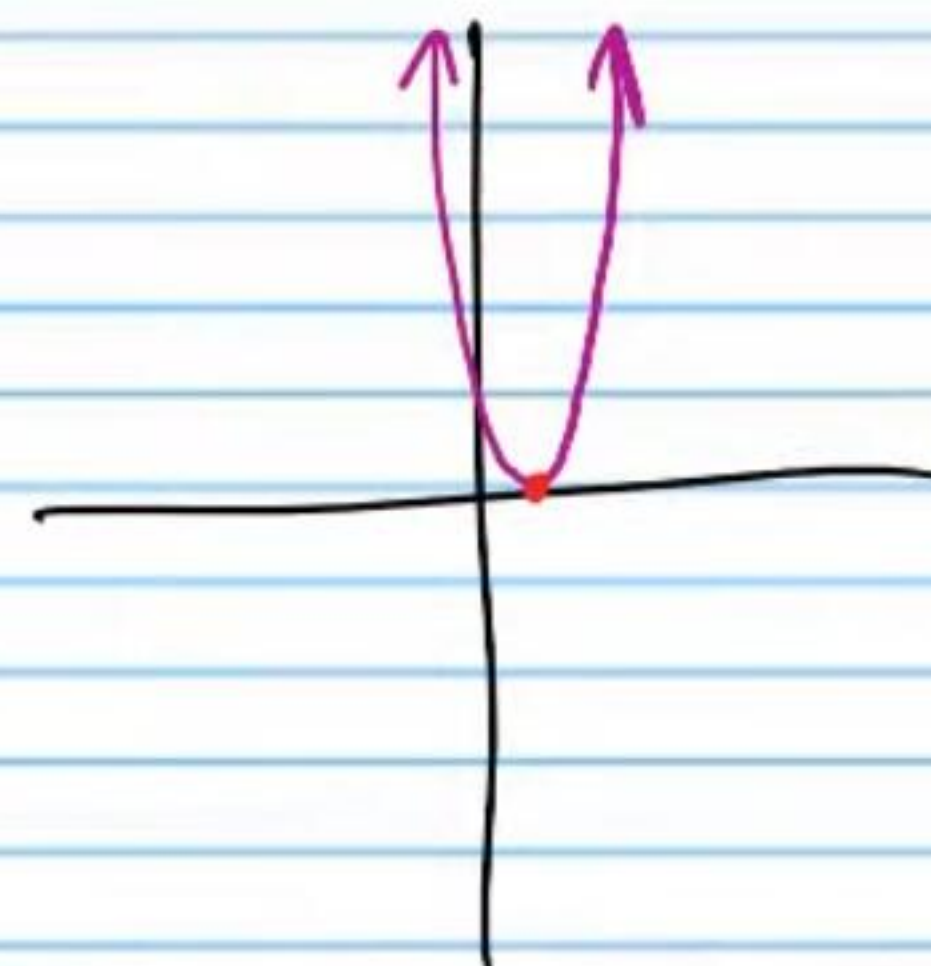
$$D = (-10)^2 - 4(25)(1)$$

$$D = 100 - 100$$

$$D = 0$$

One Solution

One X-intercept



★ You may use a calculator.

$$\textcircled{5} \ 10x^2 - 33x - 7 = y$$

$$10x^2 - 33x - 7 = 0$$

$$D = b^2 - 4ac$$

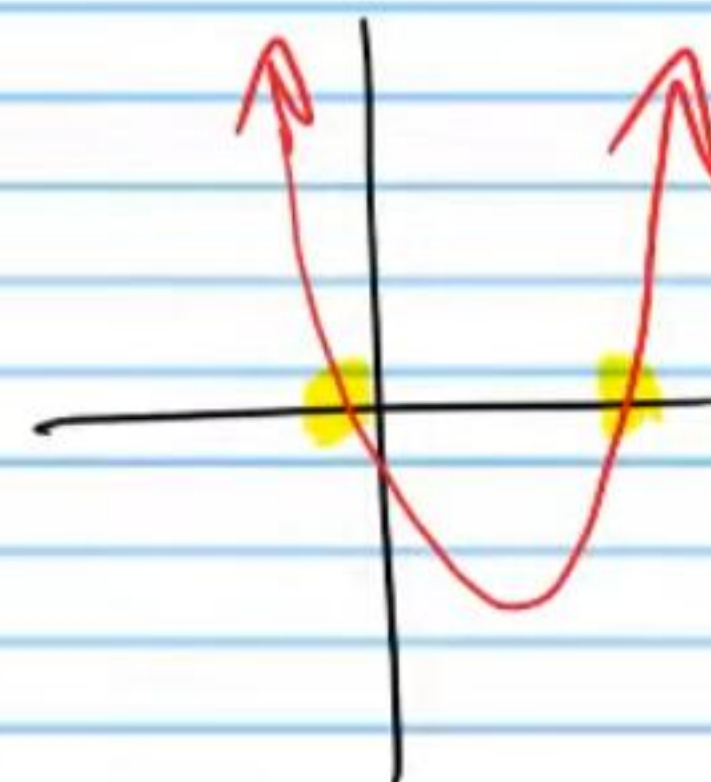
$$D = (-33)^2 - 4(10)(-7)$$

$$D = 1089 + 280$$

$$D = 1,369$$

Two Solutions

Two X-intercepts



$$⑥ y = -x^2 - 3x - 3$$

$$0 = -x^2 - 3x - 3$$

$$D = b^2 - 4ac$$

$$D = (-3)^2 - 4(-1)(-3)$$

$$D = 9 - 12$$

$$D = -3$$

No solutions

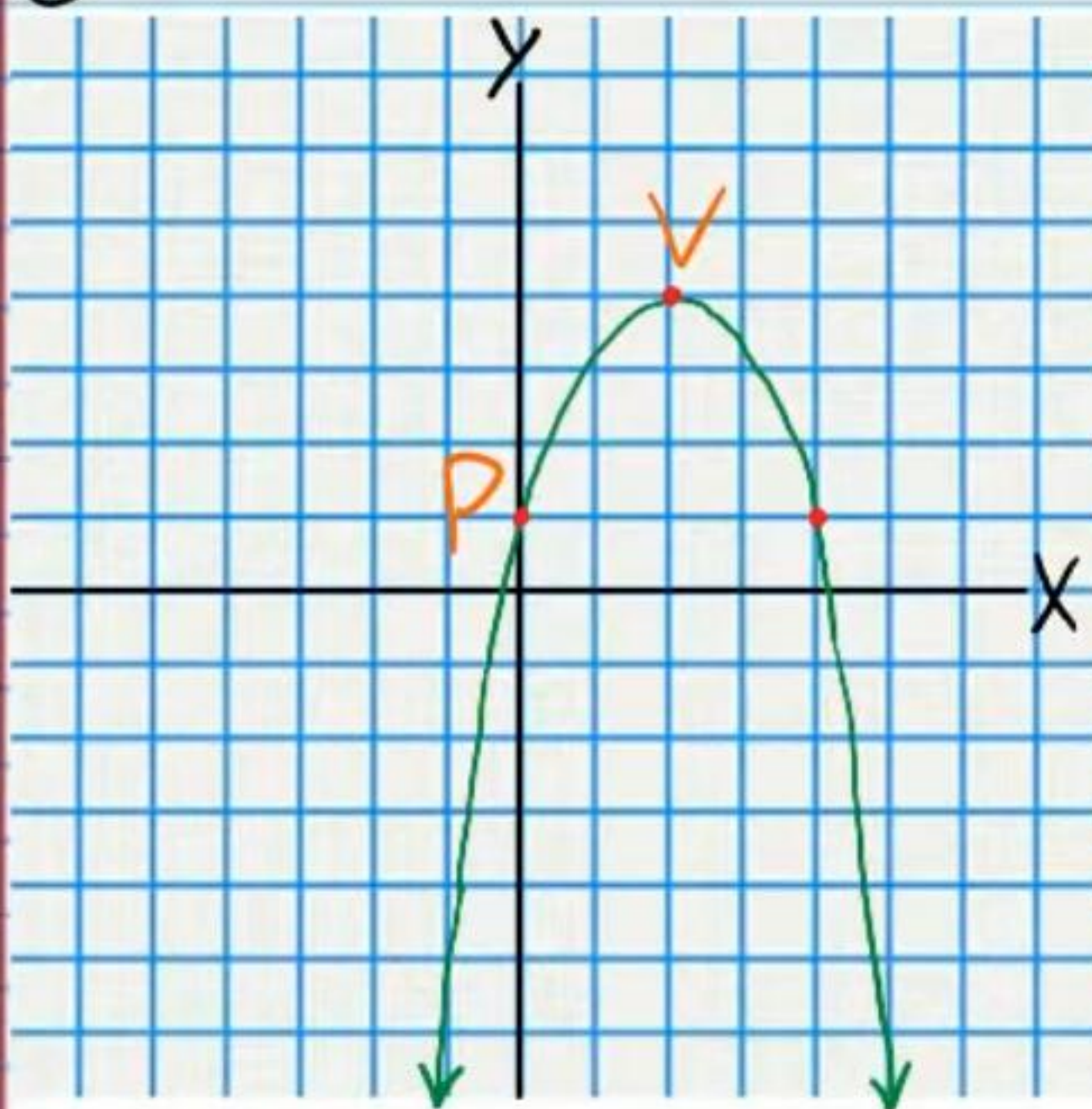
No x-intercepts



Writing Quadratic Equations Given Their Graphs

Write an equation of each parabola. Hint: It is easiest to use vertex form.

⑦



I) Find "a"

$$V(2, 4) \text{ \& \& } P(0, 1)$$

$$y = a(x-h)^2 + k$$

$$1 = a(0-2)^2 + 4$$

$$1 = a(-2)^2 + 4$$

$$1 = a(4) + 4$$

$$1 = 4a + 4$$

$$-3 = 4a$$

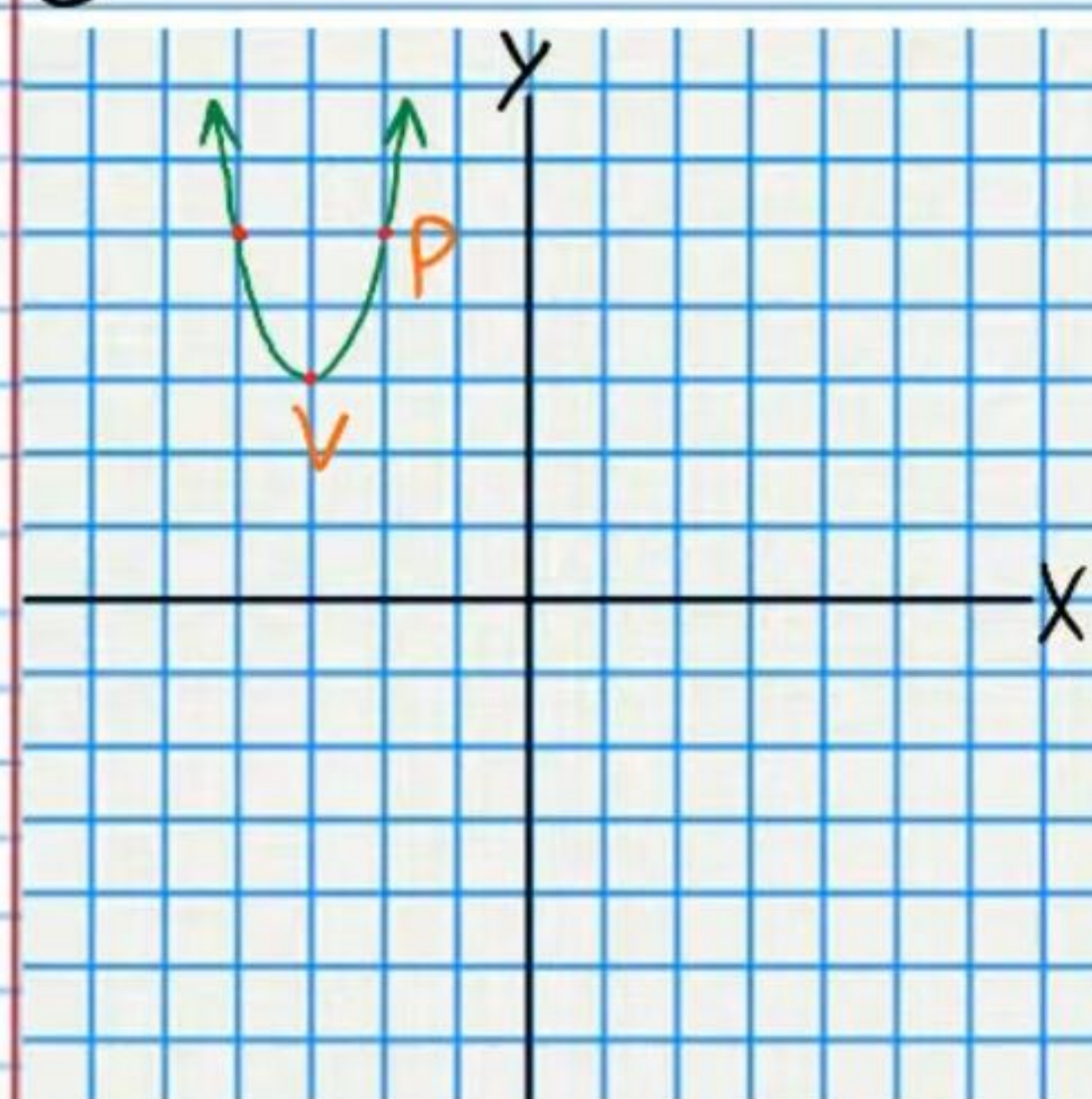
$$-\frac{3}{4} = a$$

II) Write the equation.

$$y = a(x-h)^2 + k$$

$$\boxed{y = -\frac{3}{4}(x-2)^2 + 4}$$

⑧



I) Find "a"

$$V(-3, 3) \text{ \& \& } P(-2, 5)$$

$$y = a(x-h)^2 + k$$

$$5 = a(-2-(-3))^2 + 3$$

$$5 = a(-2+3)^2 + 3$$

$$5 = a(1)^2 + 3$$

$$5 = a(1) + 3$$

$$5 = a + 3$$

$$2 = a$$

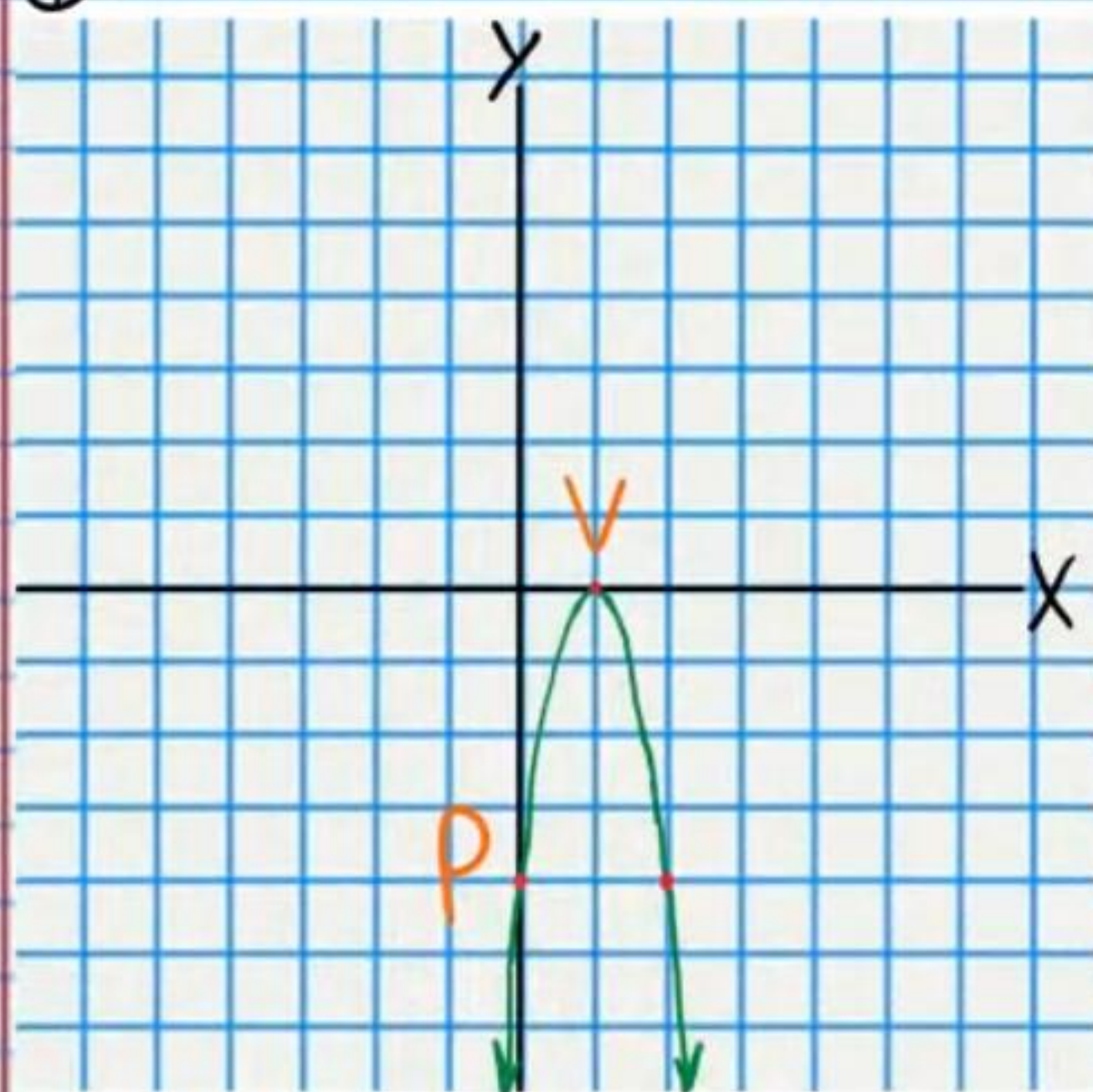
II) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = 2(x-(-3))^2 + 3$$

$$\boxed{y = 2(x+3)^2 + 3}$$

9



I) Find "a"

$$V(1, 0) \text{ \& \; } P(0, -4)$$

$$y = a(x-h)^2 + k$$

$$-4 = a(0-1)^2 + 0$$

$$-4 = a(-1)^2$$

$$-4 = a(1)$$

$$-4 = a$$

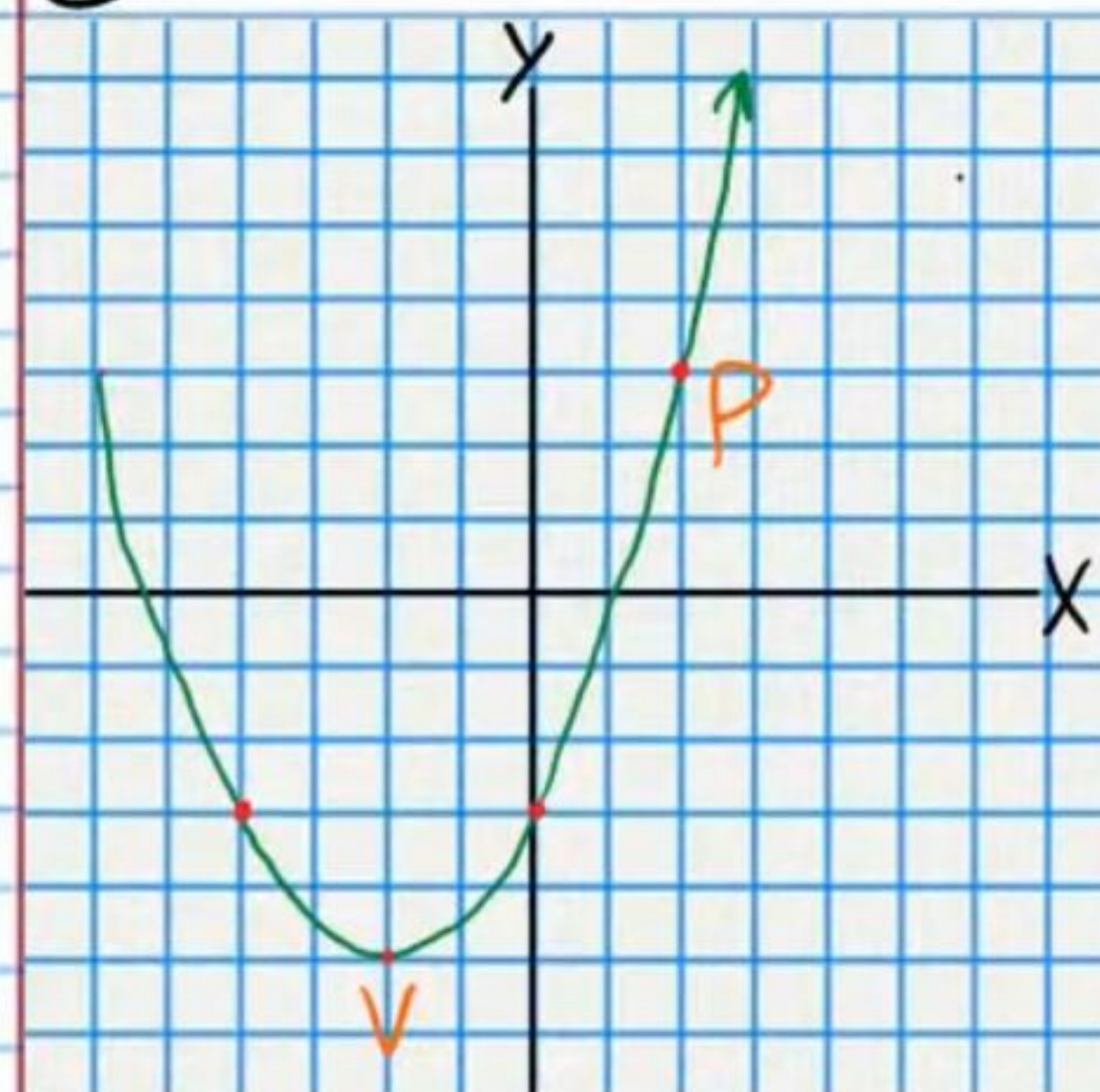
II) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = -4(x-1)^2 + 0$$

$$\boxed{y = -4(x-1)^2}$$

10



I) Find "a"

$$V(-2, -5) \text{ \& \; } P(2, 3)$$

$$y = a(x-h)^2 + k$$

$$3 = a(2-(-2))^2 - 5$$

$$3 = a(2+2)^2 - 5$$

$$3 = a(4)^2 - 5$$

$$3 = a(16) - 5$$

$$3 = 16a - 5$$

$$8 = 16a$$

$$\frac{1}{2} = a$$

II) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = \frac{1}{2}(x-(-2))^2 - 5$$

$$\boxed{y = \frac{1}{2}(x+2)^2 - 5}$$

Writing Quadratic Equations Given Information about Their Graphs

For each problem below, write an equation of the parabola with the given characteristics. *Hint: It is easiest to use vertex form. Assume that all corresponding parabolas open up or down.*

⑪ Vertex $(-4, 7)$ and point $(1, -1)$.

I) Find "a".

$$V(-4, 7) \text{ \& } P(1, -1)$$

$$y = a(x-h)^2 + k$$

$$-1 = a(1 - (-4))^2 + 7$$

$$-1 = a(1+4)^2 + 7$$

$$-1 = a(5)^2 + 7$$

$$-1 = a(25) + 7$$

$$-1 = 25a + 7$$

$$-8 = 25a$$

$$-\frac{8}{25} = a$$

II) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = -\frac{8}{25}(x - (-4))^2 + 7$$

$$y = -\frac{8}{25}(x+4)^2 + 7$$

⑫ Vertex $(2, -3)$ and point $(5, 6)$.

I) Find "a".

$$V(2, -3) \text{ \& } P(5, 6)$$

$$y = a(x-h)^2 + k$$

$$6 = a(5 - 2)^2 - 3$$

$$6 = a(3)^2 - 3$$

$$6 = a(9) - 3$$

$$6 = 9a - 3$$

$$9 = 9a$$

$$1 = a$$

II) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = (x-2)^2 - 3$$

⑬ Vertex (1,4) and x-intercept -6.

I) Find "a"

V(1,4) & P(-6,0)

$$y = a(x-h)^2 + k$$

$$0 = a(-6-1)^2 + 4$$

$$0 = a(-7)^2 + 4$$

$$0 = a(49) + 4$$

$$0 = 49a + 4$$

$$-4 = 49a$$

$$\frac{-4}{49} = a$$

II) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = -\frac{4}{49}(x-1)^2 + 4$$

⑭ Vertex (-6,2) and y-intercept 4.

I) Find "a"

V(-6,2) & P(0,4)

$$y = a(x-h)^2 + k$$

$$4 = a(0-(-6))^2 + 2$$

$$4 = a(0+6)^2 + 2$$

$$4 = a(6)^2 + 2$$

$$4 = a(36) + 2$$

$$4 = 36a + 2$$

$$2 = 36a$$

$$\frac{1}{18} = a$$

II) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = \frac{1}{18}(x-(-6))^2 + 2$$

$$y = \frac{1}{18}(x+6)^2 + 2$$

Finding the Focus and Directrix

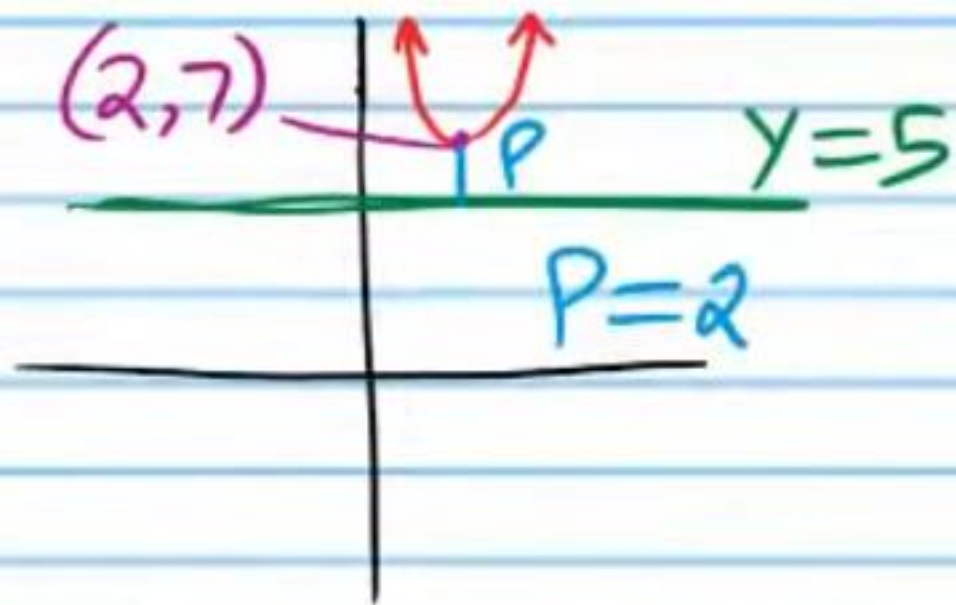
If a quadratic expression is written in any of the forms $y = ax^2 + bx + c$, $y = a(x-m)(x-n)$, or $y = a(x-h)^2 + k$, and (V_x, V_y) are the coordinates of the vertex, then the coordinates of the focus are $(V_x, V_y + P)$ and the equation of the directrix is $y = V_y - P$ if P is defined to be the expression below.

$$P = \frac{1}{4a}$$

⑮ Vertex $(2, 7)$ and directrix $y = 5$.

I) Find "P"

Conceptual Method (graphing)



Algebraic Method

$$y = V_y - P$$

$$5 = 7 - P$$

$$-2 = -P$$

$$2 = P$$

II) Find "a"

$$P = \frac{1}{4a}$$

$$2 = \frac{1}{4a}$$

$$\frac{2}{1} = \frac{1}{4a}$$

$$\frac{8a}{4a} = \frac{1}{4a}$$

$$8a = 1$$

$$a = \frac{1}{8}$$

III) Write the equation

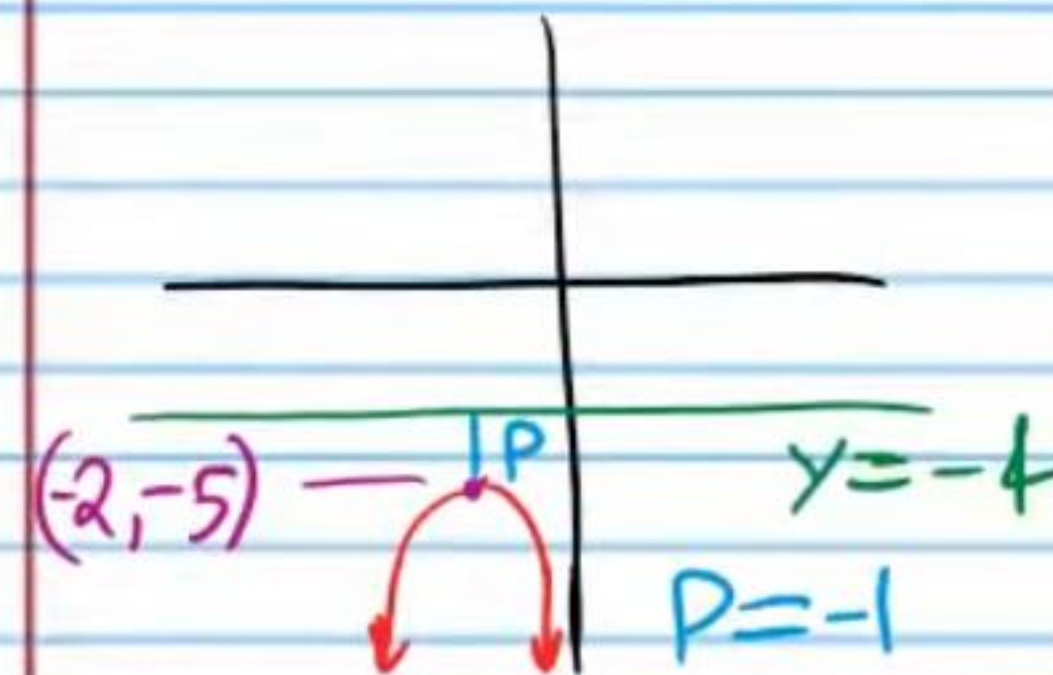
$$y = a(x-h)^2 + k$$

$$y = \frac{1}{8}(x-2)^2 + 7$$

⑯ Vertex $(-2, -5)$ and directrix $y = -4$.

I) Find "P"

Conceptual Method (graphing)



Algebraic Method

$$y = V_y - P$$

$$-4 = -5 - P$$

$$1 = -P$$

$$-1 = P$$

II) Find "a"

$$P = \frac{1}{4a}$$

$$-1 = \frac{1}{4a}$$

$$\frac{-1}{1} = \frac{1}{4a}$$

$$\frac{-4a}{4a} = \frac{1}{4a}$$

$$-4a = 1$$

$$a = -\frac{1}{4}$$

III) Write the equation.

$$y = a(x-h)^2 + k$$

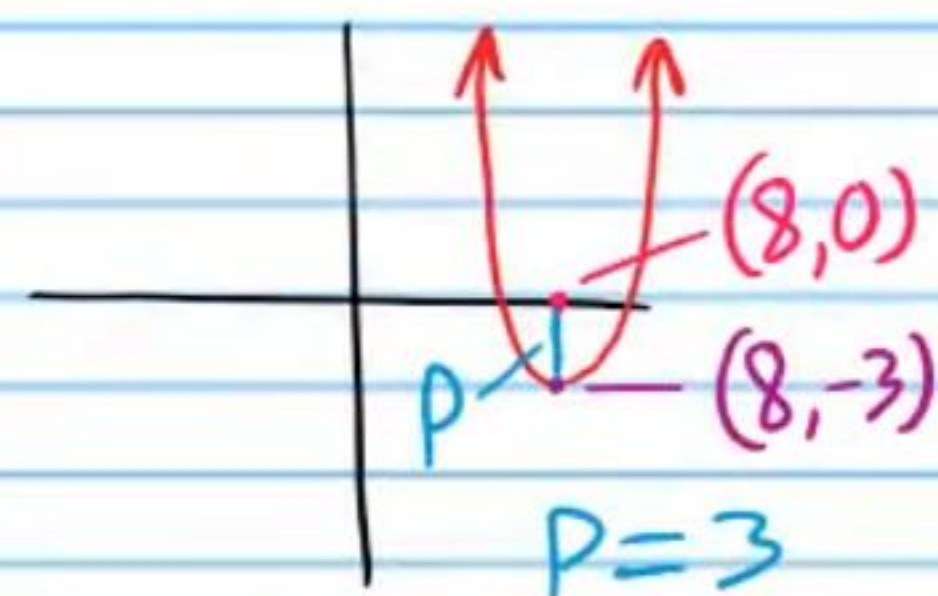
$$y = -\frac{1}{4}(x-(-2))^2 - 5$$

$$y = -\frac{1}{4}(x+2)^2 - 5$$

⑪ Vertex $(8, -3)$ and focus $(8, 0)$.

I) Find "P"

Conceptual Method (graph)



Algebraic Method

$$F(V_x, V_y + P)$$

$$F(8, 0)$$

$$V_y + P = 0$$

$$-3 + P = 0$$

$$P = 3$$

II) Find "a"

$$P = \frac{1}{4a}$$

$$3 = \frac{1}{4a}$$

$$\frac{3}{1} = \frac{1}{4a}$$

$$\frac{12a}{4a} = \frac{1}{4a}$$

$$12a = 1$$

$$a = \frac{1}{12}$$

III) Write the equation.

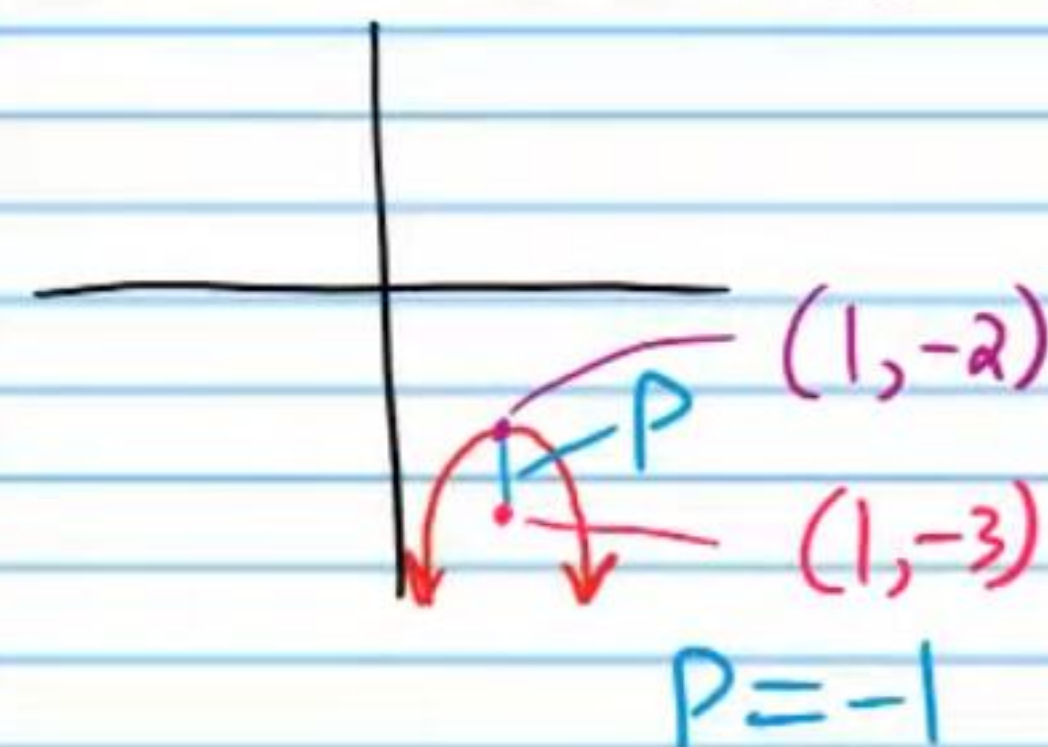
$$y = a(x-h)^2 + k$$

$$y = \frac{1}{12}(x-8)^2 - 3$$

⑫ Vertex $(1, -2)$ and focus $(1, -3)$.

I) Find "P"

Conceptual Method (graphing)



Algebraic Method

$$F(V_x, V_y + P)$$

$$F(1, -3)$$

$$V_y + P = -3$$

$$-2 + P = -3$$

$$P = -1$$

II) Find "a"

$$P = \frac{1}{4a}$$

$$-1 = \frac{1}{4a}$$

$$-\frac{1}{1} = \frac{1}{4a}$$

$$\frac{-4a}{4a} = \frac{1}{4a}$$

$$-4a = 1$$

$$a = -\frac{1}{4}$$

III) Write the equation.

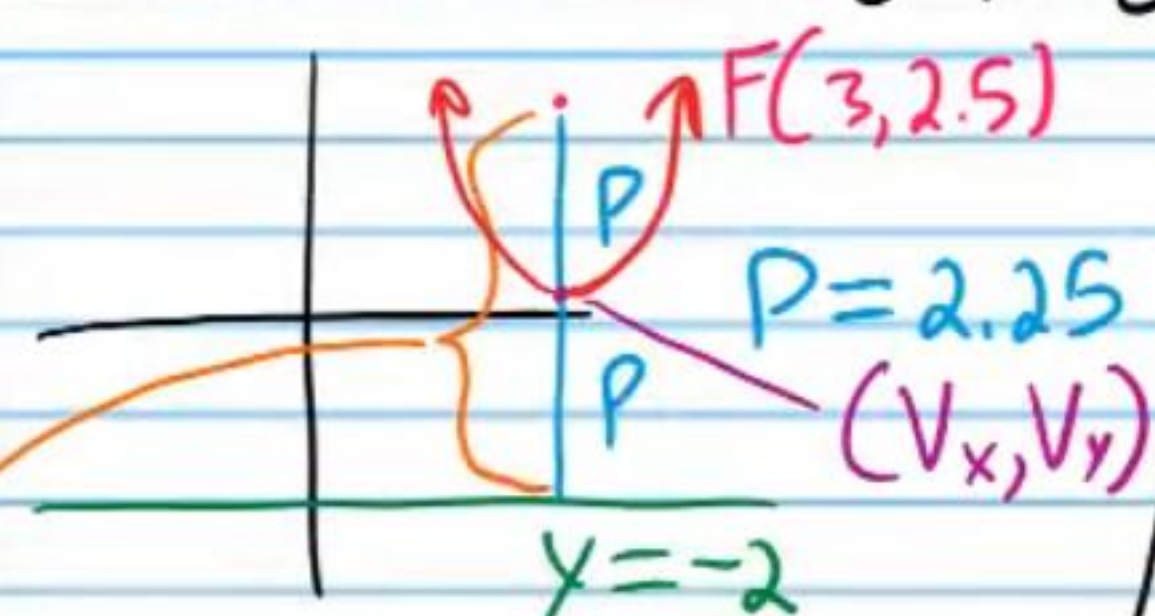
$$y = a(x-h)^2 + k$$

$$y = -\frac{1}{4}(x-1)^2 - 2$$

①9 Directrix $y = -2$ and focus $(3, 2.5)$.

I) Find "P".

Conceptual Method (graphing)



Algebraic Method

$$2P = F_y - d_y$$

$$2P = 2.5 - (-2)$$

$$2P = 2.5 + 2$$

$$2P = 4.5$$

$$P = 2.25$$

II) Find "a".

$$P = \frac{1}{4a}$$

$$2.25 = \frac{1}{4a}$$

$$\frac{2.25}{1} = \frac{1}{4a}$$

$$\frac{9a}{4a} = \frac{1}{4a}$$

$$9a = 1$$

$$a = \frac{1}{9}$$

III) Find the vertex.

$$F(V_x, V_y + P)$$

$$F(3, 2.5)$$

$$V_x = 3$$

$$V_y + P = 2.5$$

$$V_y + 2.25 = 2.5$$

$$V_y = 0.25$$

$$V_y = \frac{1}{4}$$

$$V(3, \frac{1}{4})$$

IV) Write the equation.

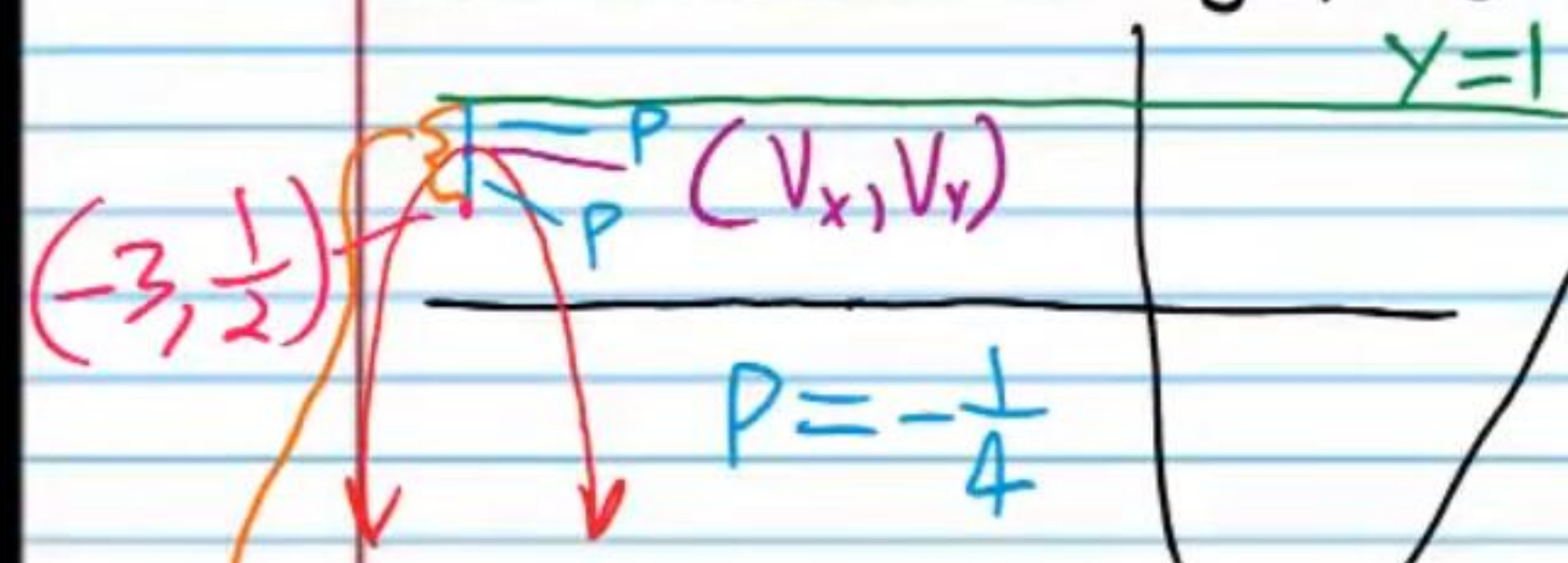
$$y = a(x-h)^2 + k$$

$$y = \frac{1}{9}(x-3)^2 + \frac{1}{4}$$

②0 Directrix $y = 1$ and focus $(-3, \frac{1}{2})$.

I) Find "P".

Conceptual Method (graphing)



Algebraic Method

$$2P = F_y - d_y$$

$$2P = \frac{1}{2} - 1$$

$$2P = -\frac{1}{2}$$

$$P = -\frac{1}{4}$$

II) Find "a".

$$P = \frac{1}{4a}$$

$$-\frac{1}{4} = \frac{1}{4a}$$

$$\frac{-a}{4a} = \frac{1}{4a}$$

$$-a = 1$$

$$a = -1$$

III) Find the vertex

$$F(V_x, V_y + P)$$

$$F(-3, \frac{1}{2})$$

$$V_y + P = \frac{1}{2}$$

$$V_y + (-\frac{1}{4}) = \frac{1}{2}$$

$$V_y = \frac{3}{4}$$

$$V_x = -3$$

$$V(-3, \frac{3}{4})$$

IV) Write the equation.

$$y = a(x-h)^2 + k$$

$$y = -(x-(-3))^2 + \frac{3}{4}$$

$$y = -(x+3)^2 + \frac{3}{4}$$